FERNANDO ARTUR AUGUSTO

AI/ML Engineer & Embedded Systems Developer
Chimoio, Moçambique · 26/11/2006

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PT: Native · EN: Intermediate · FR: Intermediate

PROFILE

Self-taught AI/ML engineer and embedded systems developer, currently completing secondary school in Mozambique. At 19, has independently built 4 production-grade AI/robotics systems using Python, C++, Arduino, and Raspberry Pi. Seeking a university scholarship to formalise and accelerate a strong foundation in applied AI, aerospace engineering, and predictive systems.

EDUCATION

Samora Machel Secondary School — Chimoio, Mozambique

- Field: CNMG (Natural Sciences, Mathematics, Geography)
- Main subjects: Chemistry, Physics, Mathematics, Geography
- Final average: 13/20

TECHNICAL PROJECTS

1. Boston Aerospace AI — Predictive Maintenance Platform

2025 · Production-grade SaaS for aeronautical engine predictive maintenance

- Ensemble of 6 ML models (XGBoost, LightGBM, CatBoost, Random Forest, GBR, Extra Trees) trained on NASA CMAPSS dataset with 18 real turbofan engine sensors
- RUL (Remaining Useful Life) prediction with 95% confidence interval and SHAP explainability — shows which sensors drove each prediction
- RAG Chat powered by Ollama (tinyllama) + ChromaDB + LangChain: indexes aeronautical maintenance manuals and answers technical questions in Portuguese
- Computer vision modules: crack detection (OpenCV + Canny), thermal analysis (JET colormap), vibration anomaly detection via MFCC audio analysis (Librosa)
- Full-stack deployment: Flask + PostgreSQL + Docker + Cloudflare Tunnel; CI/CD via GitHub Actions; PDF maintenance report export with ReportLab
- Business model developed: Free / Pro €49/month / Enterprise €199/month; market analysis covering €100B+ MRO sector

Technologies: Python, Flask, PostgreSQL, XGBoost, LightGBM, CatBoost, SHAP, LangChain, Ollama, ChromaDB, OpenCV, Librosa, Docker, GitHub Actions, Cloudflare

2. Intelligent Obstacle Detection System with Machine Learning

2024 · Embedded ML system with real-time processing

- Developed using Arduino, ultrasonic sensors, and custom ML algorithms
- Real-time data processing with 95% detection accuracy
- Visual control implemented with RGB LEDs based on adaptive safety logic

Technologies: C++, Arduino IDE, Ultrasonic Sensors, Machine Learning Logic

3. Multimodal AI Platform with Triple Processing

2024 · Integrated system processing text, images, and video simultaneously

- Intelligent Chat: Natural language processing with Google Gemini AI
- Image Module: Analysis with OpenCV, metadata extraction, aerospace applications
- Video Processing: Temporal analysis, frame extraction, integrated streaming

Technologies: Python, Streamlit, OpenCV, TensorFlow, Hugging Face, Pillow

4. Autonomous Robot with Computer Vision

2024 · Advanced development stage

- Integration of Raspberry Pi, Arduino, and multiple sensors (LiDAR, IMU, Camera)

- Autonomous navigation using OpenCV, YOLO, and TensorFlow Lite
- Voice synthesis system for real-time status reporting
- Multi-sensor data fusion for decision-making

Technologies: Python, C++, OpenCV, YOLO, ROS, TensorFlow Lite

EXPERIENCE & LEADERSHIP

Student Association — Samora Machel Secondary School

Core Member · 2022–2023

- Organisation of scientific and cultural events for 500+ students
- Representation of student interests before school administration
- Promotion of collective academic development through tutoring

SAJ Program (Youth-Friendly Service)

Volunteer · 2022–2023

- Prevention of early marriages through awareness lectures
- Promotion of healthy youth development in school settings

Peer Tutor for Students with Learning Difficulties

Volunteer · 2021–2023

- Personalised academic support for 20+ students in science subjects
- Improved overall academic performance by 15%

SKILLS

Technical

- Programming: C++, Python, Arduino, Streamlit
- AI/ML: Machine Learning, Computer Vision, NLP, SHAP
- Frameworks: OpenCV, TensorFlow, YOLO, LangChain, ChromaDB
- Backend: Flask, PostgreSQL, Docker, GitHub Actions
- Embedded: Arduino, Raspberry Pi, Sensors, Motor Control
- Tools: Microsoft Office, GitHub, Arduino IDE, Raspberry Pi OS

Interpersonal

- Complex Problem Solving and Algorithmic Thinking
- Team Leadership and Project Management
- Multilingual Communication (PT/EN/FR)
- Adaptability to New Technologies and Methodologies
- Self-directed Learning and Independent Research

KEY ACHIEVEMENTS

- Built Boston Aerospace AI — a production-grade SaaS predictive maintenance system from scratch, including ML ensemble, RAG chat, computer vision, and full cloud deployment
- Development of 4 complex AI/robotics systems through entirely self-directed learning
- Mastery of 12+ technologies (Python, C++, Docker, LangChain, ChromaDB, SHAP, etc.) through hands-on practice
- Leadership in student initiatives impacting 500+ peers
- Creation of a technical portfolio with integrated multimodal and aerospace AI projects